Chris Dai

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Education

Harvard University — Aug. 2026 – Expected May 2028
M.S. Health Data Science, T.H. Chan School of Public Health — *Boston, Massachusetts, USA*

University of Michigan, Ann Arbor — Aug. 2024 – May 2026
B.S.E. Industrial and Operations Engineering; Minor in Computer Science — *Ann Arbor, Michigan, USA*

UM–SJTU Joint Institute, Shanghai Jiao Tong University — Sep. 2022 – Aug. 2026
B.S.E. Electrical and Computer Engineering — *Shanghai, China*

Dean's Honor List (2024, 2025) · Myun Lee Scholarship (2025)

Skills

LLM & Evaluation: SFT corpus construction, synthetic dialogue generation & data flywheels, LLM-as-judge evaluation, RAG & embedding retrieval (bge-m3), prompt architecture, agent benchmarking

ML & Data: PyTorch, K-means / PCA, time-series feature engineering, SQL, PostgreSQL, dbt, evaluation-harness design

Engineering: Python, TypeScript, JavaScript, FastAPI, Flask, React, Node.js, Next.js, Playwright / Chrome DevTools Protocol

Experience

AISpeech (Capstone Sponsor) — Jun. 2026 – Aug. 2026
Sponsored Capstone Thesis, UM–SJTU Joint Institute — LLM data flywheel for in-vehicle voice assistants — *Shanghai, China*

- Led the **data flywheel** end to end for the multimedia domain of a 5-person in-vehicle voice-assistant thesis: built a **neuro-symbolic user simulator** — **41 persona archetypes** over an explicit agenda state machine — emitting **retrieval-grounded** rollouts against a **bge-m3 top-K=20** tool index.
- Scored every rollout with an **LLM-as-judge** (**397B** model) on a GPU cluster, then regenerated **200** dialogues for each of the **11 highest-error functions**; spurious no-tool calls fell **4,424→1,410**.
- Produced **17,200** verified (reward=1) multimedia-domain fine-tuning dialogues — 15,000 base plus 2,200 error-targeted — as a four-quadrant corpus (positive/negative × single/multi-turn), each batch cleared by the same automated quality gate.
- **Fine-tuning on this corpus lifted** judge-reviewed tool-call accuracy **72.8%→82.7%** on real conversations, **76.8%→98.3%** on the synthetic validated set and **83.4%→96.0%** on the user-profile set.

ByteDance — May 2026 – Aug. 2026
AI Solutions Intern; Stack: Python, Flask, React, TypeScript, Node.js, Playwright / CDP — *Shanghai, China*

- Built an **agent evaluation platform** benchmarking two enterprise AI agents on **200+** real user-task cases: drove both Electron clients over the **Chrome DevTools Protocol** and normalized their telemetry into one comparable trace, so the two could be scored side by side.
- Productized customer technical due diligence into a **citation-backed research tool** whose every conclusion resolves to its source; the regional client team adopted it as their working tool.
- Shipped a client-facing **AI demo site** — **11** interactive demos, **13** case studies — editable by non-engineers via a low-code table.

Shenzhen Legion Technology Co., Ltd. — Apr. 2025 – Present
Data Engineering & AI Intern, then Long-Term Collaborator (part-time); Stack: Python, FastAPI, PostgreSQL, React/TS — *Remote, China*

- **Co-developing MetaFlow** (two-person project, since Mar. 2026), a **metadata-centric NL-to-SQL** system that answers questions through a **verifiable relational IR** instead of generating SQL directly; I own the metadata layer, the evaluation suite and the studio front end — my collaborator owns the query engine.
- Built its **gold-case regression suite** — **20** scenarios pinned across **5** pipeline stages as **contract-as-test** — plus a **contracts endpoint** running data-quality rules as executable checks, so a metadata or schema change is caught by re-running the suite, not by hand.
- During the internship phase, designed the **prompt architecture** of an open-source **RAG data-governance copilot** for dbt pipelines and lineage — question classification (catalog / SQL / data-QA), entity extraction, embedding-based semantic retrieval — while owning feature scope and release priorities rather than the upstream codebase.
- Built that copilot's **evaluation framework** of structured test cases spanning retrieval accuracy and governance compliance, so a prompt change was measured against fixed cases instead of spot-checked.

PGF Technology Group — Mar. 2026 – Apr. 2026
University Consulting Engagement (4-person team), sole developer; Stack: Next.js, TypeScript, Postgres — *Rochester Hills, Michigan, USA*

- Sole developer of the software on a four-person consulting engagement: shipped a **full-stack work-order tracking system** covering all **9** production stations of a ∼50-person harness manufacturer — a complete solution the client took into internal use after handover.
- Wrote an **Excel/CSV bridge** (439 lines) absorbing **30+** real ERP column-name variants and sub-order merges, so exports from the client's Epicor system — only 30–40% of fields reliably filled — loaded without hand cleanup.
- Diagnosed stale reads across serverless instances and migrated storage in production (in-memory → Blob → Postgres); pinned the scan, queue and alert flows with **6 Playwright e2e suites** (582 lines).

Projects

Cardiac Rehabilitation Wearable-Data Study — Apr. 2025 – Jan. 2026
SURE Research Program, University of Michigan — Advisors: Profs. Mariel Lavieri & Jessica Golbus — *Ann Arbor, Michigan, USA*

- Built a **time-series feature pipeline** over **400K+** daily step records from **247** enrolled cardiac-rehab patients — rolling means, trend slopes, periodicity, plus imputation for gapped and high-variance series.
- Phenotyped recovery trajectories with **K-means/PCA** and detected plateaus with a **Kalman filter**: against the standard 36-session protocol, ∼**68%** of the 191-patient analysis cohort was discharged off its true plateau.

Real-Time EMG Biofeedback & Muscle-Synergy Analysis — Oct. 2024 – Apr. 2026
NeuRRo Lab, University of Michigan — PI: Chandramouli Krishnan — *Ann Arbor, Michigan, USA*

- Built the lab's real-time EMG biofeedback training system (**MATLAB**, **Unity/C#**) and its muscle-synergy analysis pipeline, both now standard tools in the lab's ongoing clinical experiments.